Exp-4:
Write a program in Dart that find the area of a circle using function.

```
import 'package:flutter/material.dart';
import 'dart:math';

void main() => runApp(const CircleAreaApp());

class CircleAreaApp extends StatelessWidget {
 const CircleAreaApp({super.key});

 @override
 Widget build(BuildContext context) {
  return MaterialApp(
   title: 'Circle Area Calculator',
   theme: ThemeData(
    primarySwatch: Colors.blue,
    scaffoldBackgroundColor: Colors.white,
    inputDecorationTheme: const InputDecorationTheme(
     border: OutlineInputBorder(),
    ),
   ),
   home: const CircleAreaScreen(),
  );
 }
}

class CircleAreaScreen extends StatefulWidget {
 const CircleAreaScreen({super.key});

 @override
 State<CircleAreaScreen> createState() => _CircleAreaScreenState();
}

class _CircleAreaScreenState extends State<CircleAreaScreen> {
 final TextEditingController _controller = TextEditingController();
 String _result = '';

 double calculateArea(double radius) => pi * radius * radius;

 void _calculate() {
  final input = _controller.text.trim();
  final radius = double.tryParse(input);
  if (radius == null || radius < 0) {
   setState(() {
    _result = 'Please enter a valid positive number';
   });
   return;
  }
```

```
  final area = calculateArea(radius);
  setState(() {
    _result = 'Area of circle: ${area.toStringAsFixed(2)}';
  });
 }

 @override
 Widget build(BuildContext context) {
  return Scaffold(
    appBar: AppBar(title: const Text('Circle Area Calculator')),
    body: Padding(
      padding: const EdgeInsets.all(24),
      child: Column(
        mainAxisAlignment: MainAxisAlignment.center,
        children: [
          TextField(
            controller: _controller,
            keyboardType:
            const TextInputType.numberWithOptions(decimal: true),
            decoration: const InputDecoration(
              labelText: 'Enter radius',
              hintText: 'e.g. 3.5',
            ),
            onSubmitted: (_) => _calculate(),
          ),
          const SizedBox(height: 20),
          ElevatedButton(
            onPressed: _calculate,
            child: const Text('Calculate'),
          ),
          const SizedBox(height: 24),
          Text(
            _result,
            style: const TextStyle(fontSize: 20, fontWeight: FontWeight.w500),
          )
        ],
      ),
    ),
  );
 }
}
```

Output: